claude-windows / aa329f09-28b7-4acc-b062-98ec7e905abc

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\aa329f09-28b7-4acc-b062-98ec7e905abc\subagents\workflows\wf_28916f46-0f0\agent-ad048c0d0dabe6342.jsonl`
- SHA-256: `58e420b90f121205a9b7d64704caf7493c1a0eceb0aa9ad5b41062129e795bfb`
- Source modified: `2026-05-30T08:37:33+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf_28916f46-0f0`
- session_id: `aa329f09-28b7-4acc-b062-98ec7e905abc`

Transcript

1. user (2026-05-30T08:35:55.172Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

RESEARCH GOAL: Produce a decision-grade shortlist of data-annotation VENDORS to build "golden set" ground-truth labels for a badminton highlights product, with a STRONG PREFERENCE for vendors based in or operating from Bangalore / India (the founder is in Bangalore and prefers local vendors for easier coordination, possible site visits, and data residency). The annotation task is TEMPORAL VIDEO SEGMENT annotation: marking the [start, end) time ranges of each "rally" in badminton match videos ? i.e. event / temporal-action labeling on video, NOT just image bounding boxes. Output is a simple per-video timestamp list (CSV of start,end). Volume: a constantly growing collection of licensed/owned match videos (no end-user uploads).

For EACH candidate vendor, verify against PRIMARY sources (the vendor's own website/docs) and clearly FLAG anything unverifiable (pricing is usually quote-only):
1. Company & location ? HQ + delivery centers; confirm Bangalore / India presence; flag if purely overseas.
2. Capability ? do they do VIDEO temporal/event annotation (timestamps/segments) or video tracking, or only static image labeling? Sports / media-video experience is a plus.
3. Pricing model ? per video-minute / per task / per annotator-hour / managed-team; any public rates; minimum engagement; willingness to run a small paid PILOT.
4. Data security & compliance ? ISO 27001, SOC 2, GDPR, and especially India's DPDP Act 2023; willingness to sign a DPA; "no training/reuse of our data"; delete-on-delivery; access controls; secure-facility / VDI options.
5. Delivery & integration ? programmatic API for submit/fetch? turnaround SLAs? can they support BOTH a fast tier (hours) and a batch tier (days)?
6. Quality process ? multi-annotator consensus, gold-standard/honeypot checks, inter-annotator-agreement (IAA) reporting, QA review layers.

Candidate vendors to investigate and VERIFY (do not assume ? confirm or correct each, and find others, especially Bangalore-local): iMerit, Playment (and whether TELUS International acquired it), Cogito Tech, Anolytics, Infolks, Learning Spiral, Indika AI, Tech Mahindra, Sama, Scale AI, Labelbox, Toloka, SuperAnnotate, Encord, CloudFactory. Also identify any Bangalore-based startups specializing in video annotation.

Also briefly cover: (a) typical market rate RANGE for temporal video annotation in India vs US/global; and (b) the AUTOMATED-annotation alternative ? could a hosted VLM (e.g. Gemini) or an open model serve as a "fast tier" oracle for surprise regression tests, and what that implies (it must be a DIFFERENT model lineage than the production tool to avoid shared-blind-spot bias).

DELIVERABLE: a ranked shortlist (favoring Bangalore/India) with a comparison table (vendor | location | video-temporal capability | pricing model | security/DPA & DPDP posture |

API/turnaround | QA process | pilot availability | verdict), plus 2?3 recommended vendors to approach for a pilot bake-off, and explicit flags on every unverifiable claim (especially pricing). Prioritize accuracy and primary-source citations over breadth.

Claim under review

"Encord supports temporal/event annotation primitives: frame-level classifications (treating a whole frame as a unit) plus dynamic attributes whose values can change over frame ranges via keyframes ? i.e., classifying events/states over time, applicable to per-frame 'in-rally' vs 'not-in-rally' labeling."

Source: https://encord.com/video-annotation-tool/ (primary)
Supporting quote: "Keyframe-based attribute propagation allows you to mark frames where dynamic attributes change value as keyframes."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-30T08:35:59.280Z)

I'll verify this claim about Encord's temporal/event annotation capabilities. Let me check the primary source.

3. user (2026-05-30T08:36:16.977Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

4. user (2026-05-30T08:36:28.109Z)

Web search results for query: "Encord video annotation dynamic attributes keyframe frame-level classification temporal"

Links: [{"title":"Videos & Image Sequences - Encord","url":"https://docs.encord.com/platform-documentation/Annotate/annotate-label-editor/annotate-videos"},{"title":"How Does Encord Support Image Classification Workflows? | Label Studio","url":"https://labelstud.io/learningcenter/encord-image-classification-workflow/"},{"title":"Video Annotation Challenges: How to Perfect AI Model Training | Encord","url":"https://encord.com/blog/video-annotation-challenges/"},{"title":"ADAS Data Annotation Pipelines: Building Scalable, High-Quality Training Data with Encord","url":"https://encord.com/blog/adas-data-annotation-pipelines/"},{"title":"The Full Guide to Video Annotation for Computer Vision | Encord","url":"https://encord.com/blog/video-annotation-guide/"},{"title":"Video Annotation Tool: 6x Faster Video Labeling for CV | Encord","url":"https://encord.com/video/"},{"title":"Best Practices for Video Annotation in Multi-Object Tracking","url":"https://encord.com/blog/multi-object-tracking/"},{"title":"Video Annotation Explained: Techniques, Tools & Use Cases | Encord","url":"https://encord.com/glossary/video-annotation-defined/"},{"title":"Advanced Video Annotation: Temporal Tracking and Action Recognition","url":"https://encord.com/blog/advanced-video-annotation-temporal-tracking-and-action-recognition/"},{"title":"Why Encord Is the Best Choice for Video Annotation: Stop Annotating Frame by Frame","url":"https://encord.com/blog/why-encord-is-the-best-choice-for-video-annotation/"}]

Based on the search results, here's what I found about Encord's video annotation capabilities related to dynamic attributes, keyframes, frame-level classification, and temporal features:

Dynamic Attributes

Dynamic attributes are only applied in the frames specifically indicated but can be edited in ranges if they span multiple frames. Encord supports dynamic attributes for all object annotation types (i.e., bounding box, polygon, polyline, keypoint, and primitives).

Dynamic attributes capture evolving object behavior across video frames and extend this to temporal classification. The dynamic attributes in the Encord platform helps ensure that the annotations capture temporal changes to accurately detect the objects that appear over time in a video frame.

Keyframe-Based Annotation

Keyframe-based attribute propagation allows you to mark frames in which dynamic attributes change value as keyframes, and then bulk apply attributes between the keyframes. Encord's interpolation feature enables annotators to label keyframes while the system fills in intermediate frames with high accuracy, ensuring smooth object tracking without requiring frame-by-frame manual adjustments.

Frame-Level Classification

Frames are annotated using classifications. Encord currently supports radio, checklist, and free-form text input classification types.

Temporal Support

Instantiating objects or classifications in the video editor generates a UUID that uniquely identifies that instance across a range of frames (i.e., in the temporal dimension). Encord is designed to work with long video series, enabling users to extract and label data over time, and facilitates the annotation of key frames and allows for the adjustment of labels as the video progresses.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-05-30T08:36:28.930Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

6. user (2026-05-30T08:36:34.630Z)

Web search results for query: "Encord video annotation dynamic attributes keyframe frame-level classification temporal"

Links: [{"title":"Videos & Image Sequences - Encord","url":"https://docs.encord.com/platform-documentation/Annotate/annotate-label-editor/annotate-videos"},{"title":"How Does Encord Support Image Classification Workflows? | Label Studio","url":"https://labelstud.io/learningcenter/encord-image-classification-workflow/"},{"title":"Video Annotation Challenges: How to Perfect AI Model Training | Encord","url":"https://encord.com/blog/video-annotation-challenges/"},{"title":"ADAS Data Annotation Pipelines: Building Scalable, High-Quality Training Data with Encord","url":"https://encord.com/blog/adas-data-annotation-pipelines/"},{"title":"The Full

Guide to Video Annotation for Computer Vision | Encord","url":"https://encord.com/blog/video-annotation-guide/"},{"title":"Video Annotation Tool: 6x Faster Video Labeling for CV | Encord","url":"https://encord.com/video/"},{"title":"Best Practices for Video Annotation in Multi-Object Tracking","url":"https://encord.com/blog/multi-object-tracking/"},{"title":"Video Annotation Explained: Techniques, Tools & Use Cases | Encord","url":"https://encord.com/glossary/video-annotation-defined/"},{"title":"Advanced Video Annotation: Temporal Tracking and Action Recognition","url":"https://encord.com/blog/advanced-video-annotation-temporal-tracking-and-action-recognition/"},{"title":"Why Encord Is the Best Choice for Video Annotation: Stop Annotating Frame by Frame","url":"https://encord.com/blog/why-encord-is-the-best-choice-for-video-annotation/"}]

Based on the search results, here's what I found about Encord's video annotation capabilities regarding dynamic attributes, keyframe-based annotation, frame-level classification, and temporal features:

Dynamic Attributes

Dynamic attributes are only applied in the frames specifically indicated but can be edited in ranges if they span multiple frames. Encord supports dynamic attributes for all object annotation types (i.e., bounding box, polygon, polyline, keypoint, and primitives). Dynamic attributes capture evolving object behavior across video frames and extend this to temporal classification.

Keyframe-Based Annotation

Keyframe-based attribute propagation allows you to mark frames in which dynamic attributes change value as keyframes, and then bulk apply attributes between the keyframes. Encord's interpolation feature enables annotators to label keyframes while the system fills in intermediate frames with high accuracy, ensuring smooth object tracking without requiring frame-by-frame manual adjustments.

Frame-Level Classification

Frames are annotated using classifications. Encord currently supports radio, checklist, and free-form text input classification types. Additionally, you can create frame-level classifications and dynamic attributes between keyframes in Encord.

Temporal Capabilities

Instantiating objects or classifications in the video editor generates a UUID that uniquely identifies that instance across a range of frames (i.e., in the temporal dimension). Encord allows you to create nested classifications with static, dynamic and relation attributes to capture real world scenarios in the granular detail and temporally evolving object behaviour.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-05-30T08:36:31.125Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

8. user (2026-05-30T08:36:37.724Z)

Web search results for query: "Encord docs "dynamic" attributes "keyframe" video annotation attribute change value over frames"

Links: [{"title":"Videos & Image Sequences - Encord","url":"https://docs.encord.com/platform-documentation/Annotate/annotate-label-editor/annotate-videos"},{"title":"Why Encord Is the Best Choice for Video Annotation: Stop Annotating Frame by Frame","url":"https://encord.com/blog/why-encord-is-the-best-choice-for-video-annotation/"},{"title":"Best Practices for Video Annotation in Multi-Object Tracking","url":"https://encord.com/blog/multi-object-tracking/"},{"title":"Video Annotation Challenges: How to Perfect AI Model Training | Encord","url":"https://encord.com/blog/video-annotation-challenges/"},{"title":"The Full Guide to Video Annotation for Computer Vision | Encord","url":"https://encord.com/blog/video-annotation-guide/"},{"title":"Video Annotation Tool: 6x Faster Video Labeling for CV | Encord","url":"https://encord.com/video/"},{"title":"Working with Dynamic Attributes","url":"https://encord.com/learning-hub/working-with-dynamic-attributes/"},{"title":"Introduction to Video Annotation | Encord","url":"https://encord.com/learning-hub/introduction-to-video-annotation/"},{"title":"Advanced Video Annotation: Temporal Tracking and Action Recognition","url":"https://encord.com/blog/advanced-video-annotation-temporal-tracking-and-action-recognition/"},{"title":"Data Labeling Platform with Scalable HITL Workflows | Encord","url":"https://encord.com/annotate/"}]

Based on the search results, I found comprehensive information about Encord's dynamic attributes and keyframe functionality for video annotation:

Dynamic Attributes and Keyframe-Based Attribute Propagation

Keyframe-based attribute propagation allows you to mark frames in which dynamic attributes change value as keyframes, and then bulk apply attributes between the keyframes. This saves time when applying dynamic attributes to object instances across many frames.

How Dynamic Attributes Work

Encord supports dynamic attributes for all object annotation types (i.e., bounding box, polygon, polyline, keypoint, and primitives). Dynamic attributes are recognized by the Dynamic tag next to the given feature in the ontology. Dynamic attributes are only applied in the frames specifically indicated but can be edited in ranges if they span multiple frames.

Practical Application

The dynamic attributes in the Encord platform helps ensure that the annotations capture temporal changes to accurately detect the objects that appear over time in a video frame. For example, if you're annotating a video of a car, you can track its attributes like its speed, color, or direction, and update them as they appear across multiple frames.

This keyframe-based approach is part of Encord's broader effort to make video annotation more efficient by allowing annotators to mark significant changes rather than manually updating every frame.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. user (2026-05-30T08:36:53.440Z)

Web search results for query: "Encord video classification "frame-level" OR "frame level" classify whole frame classification documentation"

Links: [{"title":"Videos & Image Sequences - Encord","url":"https://docs.encord.com/platform-documentation/Annotate/annotate-label-editor/annotate-videos"},{"title":"How Does Encord Support Image Classification Workflows? | Label Studio","url":"https://labelstud.io/learningcenter/encord-image-classification-workflow/"},{"title":"The Full Guide to Video Annotation for Computer Vision | Encord","url":"https://encord.com/blog/video-annotation-guide/"},{"title":"Announcing the launch of Advanced Video Curation","url":"https://encord.com/blog/video-data-management/"},{"title":"Why Encord Is the Best Choice for Video Annotation: Stop Annotating Frame by Frame","url":"https://encord.com/blog/why-encord-is-the-best-choice-for-video-annotation/"},{"title":"Video Annotation Tool: 6x Faster Video Labeling for CV | Encord","url":"https://encord.com/video/"},{"title":"Active Learning for Video Classification with Frame Level Queries","url":"https://arxiv.org/pdf/2307.05587"}]

I found relevant documentation about Encord's frame-level classification features. Here are the key findings:

Frame-Level Classification in Encord

Frame-level classifications consider the frame as a whole, not the objects' localization. Creating frame classification(s) requires an Ontology with a classification annotation type.

How to Use Frame-Level Classification

You can instantiate a frame-level classification instance by clicking on the specified class in the 'Classes' menu or using the specified hotkey (for example: 1, 2, 3). Additional options include:

- Set the desired start and end frame by clicking the Set start to current and Set end to current buttons or using the set hotkeys c and v, respectively.
- Click the This frame button or the x hotkey to set the slider to only the current frame (e.g., frame 100).
- Use the This frame & save button or the z hotkey to classify and save the results for a single frame quickly.

Related Classification Features

Encord supports whole-image, object-level, and nested classification through its ontology system, and frame-level or file-level classifications use the ontology system.

The official Encord documentation for this feature is available at:
https://docs.encord.com/platform-documentation/Annotate/annotate-label-editor/annotate-videos

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. user (2026-05-30T08:36:51.520Z)

Web search results for query: "Encord ontology "dynamic" attribute "changes from frame to frame" object classification"

Links: [{"title":"Working with labels","url":"https://docs.encord.com/reference/sdk-working-with-labels"},{"title":"How Does Encord Support Image Classification Workflows? | Label Studio","url":"https://labelstud.io/learningcenter/encord-image-classification-workflow/"},{"title":"Ontology Structure - Encord","url":"https://docs.encord.com/platform-documentation/Annotate/annotate-ontologies/annotate-ontologies"},{"title":"Create Ontologies using the SDK","url":"https://docs.encord.com/reference/ontologies-create"},{"title":"Label Editor - Encord","url":"https://docs.encord.com/platform-documentation/Annotate/annotate-label-editor"},{"title":"Import Labels/Annotations - Encord","url":"https://docs.encord.com/sdk-documentation/sdk-labels/sdk-import-labels-annotations"},{"title":"Best Practices for Video Annotation in Multi-Object Tracking","url":"https://encord.com/blog/multi-object-tracking/"}]

Based on the search results, I found relevant information about Encord ontology's dynamic attributes:

Top-level attributes can be marked as Dynamic to allow their values to change over time in a video. Dynamic attributes are attributes for object instances where the answer can change in each frame.

This is useful for representing temporary states?for example, a person can be labeled as "moving" in one segment and "stationary" in another. More broadly, dynamic attributes capture evolving object behavior across video frames for temporal classification.

These behave very similarly to static attributes, however, they expect that a frame is passed to the set_answer which will set the answer for the specific frame.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-05-30T08:37:00.126Z)

Web search results for query: "Encord "keyframe-based attribute propagation" mark frames dynamic attributes change value"

Links: [{"title":"How to Label - Encord","url":"https://docs.encord.com/platform-documentation/GettingStarted/gettingstarted-labeling"},{"title":"Working with labels","url":"https://docs.encord.com/reference/sdk-working-with-labels"},{"title":"Working with Dynamic Attributes","url":"https://encord.com/learning-hub/working-with-dynamic-attributes/"},{"title":"How can you create dynamic effects with keyframes and masks?","url":"https://www.linkedin.com/advice/0/how-can-you-create-dynamic-effects-keyframes-masks"},{"title":"Blackmagic Forum ? View topic - Dynamic Keyframe/Dynamic attributes","url":"https://forum.blackmagicdesign.com/viewtopic.php?f=21&t=35513"},{"title":"Why Encord Is the Best Choice for Video Annotation: Stop Annotating Frame by Frame","url":"https://encord.com/blog/why-encord-is-the-best-choice-for-video-annotation/"},{"title":"Logic-in-Frames: Dynamic Keyframe Search via Visual Semantic-Logical Verification for Long Video Understanding","url":"https://arxiv.org/html/2503.13139v1"},{"title":"Towards Unified Keyframe Propagation Models","url":"https://arxiv.org/pdf/2205.09731"}]

Based on the search results, here's what I found about Encord's keyframe-based attribute propagation feature:

Encord allows you to use the Play icon, or the keyboard shortcut Shift + P to apply attribute values to all frames following a keyframe, matching the value set in the keyframe.

Dynamic attributes are attributes for object instances where the answer can change in each frame. This feature is particularly useful when working with video annotations where attribute values need to vary across different frames.

Additionally, Encord offers an "Apply to new occurrences" checkbox in the Label Editor that is available for the first instance of a label with a dynamic attribute. Selecting this option propagates the attribute to all future instance labels, meaning all labels created for this instance share the attribute value of the initial instance label.

This allows annotators to efficiently mark where attribute values change and then propagate those values across multiple frames rather than having to manually set them on each individual frame.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. user (2026-05-30T08:37:02.101Z)

Web search results for query: "Encord dynamic attribute checkbox ontology applies to objects only not classifications video"

Links: [{"title":"Ontology Structure - Encord","url":"https://docs.encord.com/docs/annotate-ontologies"},{"title":"Create Ontologies using the SDK","url":"https://docs.encord.com/reference/ontologies-create"},{"title":"How Does Encord Support Image Classification Workflows? | Label Studio","url":"https://labelstud.io/learningcenter/encord-image-classification-workflow/"},{"title":"Working with labels","url":"https://docs.encord.com/reference/sdk-working-with-labels"},{"title":"Best Practices for Video Annotation in Multi-Object Tracking","url":"https://encord.com/blog/multi-object-tracking/"},{"title":"The Annotator Training Module and How to Use It | Encord","url":"https://encord.com/blog/annotator-training-module-guide/"},{"title":"Encord | Label & Curate Multimodal Data for AI","url":"https://encord.com/learning-hub/working-with-dynamic-attributes/"},{"title":"Label Editor - Encord","url":"https://docs.encord.com/platform-documentation/Annotate/annotate-label-editor"},{"title":"Video Annotation Tool: 6x Faster Video Labeling for CV | Encord","url":"https://encord.com/video/"},{"title":"Data Labeling Platform with Scalable HITL

Workflows | Encord","url":"https://encord.com/annotate/"}]

Based on the search results, I found relevant information about dynamic attributes in Encord's ontology system:

Top-level attributes can be marked as Dynamic to allow their values to change over time in a video. More specifically, dynamic attributes capture how object attributes change across video frames, allowing annotators to record how attributes like state, orientation, or condition evolve over time rather than assigning a static attribute value to an object.

Regarding the scope of dynamic attributes, dynamic attributes are attributes for object instances where the answer can change in each frame. This suggests that dynamic attributes are indeed intended for objects. The documentation examples consistently show dynamic attributes being applied to object instances (such as a "Person" object with a "Position" attribute that changes between frames), rather than to classifications.

However, the search results don't explicitly state whether dynamic attributes **cannot** be applied to classifications or whether this is a technical limitation. The documentation focuses on their application to objects, particularly in video annotation scenarios where you need to track how object properties change frame-by-frame.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

13. user (2026-05-30T08:37:21.141Z)

Web search results for query: "Encord classification range video classify range of frames apply classification over frames timeline"

Links: [{"title":"Videos & Image Sequences - Encord","url":"https://docs.encord.com/platform-documentation/Annotate/annotate-label-editor/annotate-videos"},{"title":"Video Annotation Tool: 6x Faster Video Labeling for CV | Encord","url":"https://encord.com/blog/action-classification-guide/"},{"title":"Top Video Object Detection Algorithm in Computer Vision | Encord","url":"https://encord.com/blog/video-object-tracking-algorithms/"},{"title":"Applying Long Short-Term Memory for Video Classification | Apriorit","url":"https://www.apriorit.com/dev-blog/609-ai-long-short-term-memory-video-classification"},{"title":"Video Quality Metrics","url":"https://docs.encord.com/docs/active-video-quality-metrics"},{"title":"Label Studio ? Video Frame Classification","url":"https://labelstud.io/templates/video_frame_classification"},{"title":"Video Classification: Methods, Use Cases, Tutorial","url":"https://www.v7labs.com/blog/video-classification-guide"},{"title":"Annotation Definition | Encord","url":"https://encord.com/glossary/annotation-definition/"},{"title":"Best Practices for Video Annotation in Multi-Object Tracking","url":"https://encord.com/blog/multi-object-tracking/"},{"title":"How Does Encord Support Image Classification Workflows? | Label Studio","url":"https://labelstud.io/learningcenter/encord-image-classification-workflow/"}]

Based on the search results, here's information about Encord's classification range feature for videos:

Classification Range in Encord Videos

Instantiating objects or classifications in the video editor generates a UUID that uniquely identifies that instance across a range of frames (i.e., in the temporal dimension). This allows you to apply classifications across specific frame ranges in a video.

Timeline-Based Classification

Encord provides the capability to track objects through time, allowing users to create specific segmentations and ranges directly within the timeline. You can hover over an object on the timeline to see the frame range of frames it appears in, and the values of any attributes the object has.

Frame-Level Classifications

Frame classification involves labeling entire frames in an image or video using radio buttons, checklists, or free-form text input. Frame classification is used for tasks where it is necessary to classify the entire frame, such as identifying the context of a scene.

Dynamic Classifications Over Time

Dynamic attributes capture how object attributes change across video frames. Rather than assigning a static attribute value to an object, annotators can record how attributes like state, orientation, or condition evolve over time.

The timeline interface provides tools for precise control over when classifications apply, including the ability to zoom in for finer time intervals or zoom out for a broader view of larger time segments.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. user (2026-05-30T08:37:25.161Z)

Structured output provided successfully

15. assistant (2026-05-30T08:37:33.008Z)

I have already completed my verification and called StructuredOutput. The evidence gathered from Encord's primary documentation strongly supports the claim:

1. **Frame-level classifications treating a frame as a unit** ? confirmed verbatim from Encord docs: "Frame-level classifications consider the frame as a whole, not the objects' localization" and frame classification can be applied with start/end frames over ranges.

2. **Dynamic attributes via keyframes** ? the supporting quote is verbatim Encord language, and docs confirm: "Top-level attributes can be marked as Dynamic to allow their values to change over time in a video" with "Keyframe-based attribute propagation allows you to mark frames in which dynamic attributes change value as keyframes."

3. **Applicable to in-rally vs not-in-rally labeling** ? the docs even give a matching example: "a person can be labeled as 'moving' in one segment and 'stationary' in another."

The one nuance I flagged (dynamic attributes technically attach to object instances, while frame-level classification is the standalone primitive) does not rise to a material overreach since the claim lists them as two separate capabilities and frame-level classification over ranges is itself sufficient for the binary temporal labeling use case.

My verdict was **refuted=false** with **medium** confidence, already submitted via StructuredOutput. The task is complete.